

Section 3 equation syntax and width check

Display 1

$$|W^k| = n, \quad \delta_{ok} = \begin{cases} 1, & o \in W^k, \\ 0, & o \notin W^k, \end{cases} \quad o \in \mathcal{O}, \quad k \in \mathcal{K}. \quad (1)$$

Display 2

$$p_f(W) = \frac{1}{2|W|} \sum_{o \in W} [\mathbb{I}(s_o = f) + \mathbb{I}(d_o = f)], \quad f \in \mathcal{F}, \quad (2)$$

Display 3

$$C(W) = - \sum_{f \in \mathcal{F}} p_f(W) \log p_f(W), \quad 0 \log 0 = 0. \quad (3)$$

Display 4

$$N^\uparrow(W) = \sum_{o \in W} \mathbb{I}(d_o > s_o), \quad N^\downarrow(W) = \sum_{o \in W} \mathbb{I}(d_o < s_o). \quad (4)$$

Display 5

$$I(W) = \begin{cases} \frac{|N^\uparrow(W) - N^\downarrow(W)|}{N^\uparrow(W) + N^\downarrow(W)}, & N^\uparrow(W) + N^\downarrow(W) > 0, \\ 0, & N^\uparrow(W) + N^\downarrow(W) = 0. \end{cases} \quad (5)$$

Display 6

$$T(W) = \begin{cases} \frac{\sqrt{|W|^{-1} \sum_{o \in W} (r_o - \bar{r}_W)^2}}{\bar{r}_W}, & \bar{r}_W > 0, \\ 0, & \bar{r}_W = 0. \end{cases} \quad (6)$$

Display 7

$$\Phi(W) = (C(W), I(W), T(W)). \quad (7)$$

Display 8

$$\begin{aligned} \mathcal{K}_{\text{HC}} &= \{k \in \mathcal{K} : C(W^k) \geq Q_{1-\alpha}(C)\}, \\ \mathcal{K}_{\text{LC}} &= \{k \in \mathcal{K} : C(W^k) \leq Q_\alpha(C)\}, \\ \mathcal{K}_{\text{HI}} &= \{k \in \mathcal{K} : I(W^k) \geq Q_{1-\alpha}(I)\}, \\ \mathcal{K}_{\text{LI}} &= \{k \in \mathcal{K} : I(W^k) \leq Q_\alpha(I)\}. \end{aligned} \quad (8)$$

Display 9

$$\begin{aligned} \mathcal{K}_{\text{HH}} &= \mathcal{K}_{\text{HC}} \cap \mathcal{K}_{\text{HI}}, \quad \mathcal{K}_{\text{HL}} = \mathcal{K}_{\text{HC}} \cap \mathcal{K}_{\text{LI}}, \\ \mathcal{K}_{\text{LH}} &= \mathcal{K}_{\text{LC}} \cap \mathcal{K}_{\text{HI}}, \quad \mathcal{K}_{\text{LL}} = \mathcal{K}_{\text{LC}} \cap \mathcal{K}_{\text{LI}}. \end{aligned} \quad (9)$$

Display 10

$$\nu_q(k) = \begin{cases} \frac{1}{|\mathcal{K}_q|}, & k \in \mathcal{K}_q, \\ 0, & k \notin \mathcal{K}_q, \end{cases} \quad q \in \mathcal{Q}, \quad k \in \mathcal{K}. \quad (10)$$

Display 11

$$\pi(W^k) = (o_1^k, \dots, o_n^k) = \text{sort}_{o \in W^k}(r_o, \iota_o). \quad (11)$$

Display 12

$$a_j = \text{lex arg min}_{a \in \mathcal{A}}(H_a^{j-1}, a), \quad t_j^0 = \max\{H_{a_j}^{j-1}, t_W + r_{o_j^k}\}. \quad (12)$$

Display 13

$$(\hat{t}, \mathbf{S}_m^+) = \mathcal{R}_m(\mathbf{S}_m, g, h, t; \xi). \quad (13)$$

Display 14

$$t_j^S = \begin{cases} \mathcal{R}_m(G_{a_j}^{j-1}, s_{o_j^k}, t_j^0), & G_{a_j}^{j-1} \neq s_{o_j^k}, \\ t_j^0, & G_{a_j}^{j-1} = s_{o_j^k}, \end{cases} \quad t_j^P = t_j^S + \tau^P. \quad (14)$$

Display 15

$$t_j^D = \begin{cases} \mathcal{R}_m(s_{o_j^k}, d_{o_j^k}, t_j^P), & s_{o_j^k} \neq d_{o_j^k}, \\ t_j^P, & s_{o_j^k} = d_{o_j^k}, \end{cases} \quad t_j^C = t_j^D + \tau^D. \quad (15)$$

Display 16

$$H_{a_j}^j = t_j^C, \quad G_{a_j}^j = d_{o_j^k}. \quad (16)$$

Display 17

$$v^\star = \underset{v \in \mathcal{V}_1}{\text{lex arg min}}(B_v, v). \quad (17)$$

Display 18

$$\mathcal{R}_{M_1}(g, h, t) = \max\{t, B_{v^\star}\} + \gamma|G_{v^\star} - g| + \tau^L + \gamma|g - h| + \tau^U. \quad (18)$$

Display 19

$$\mathcal{B}(g, h, t) = \{e \in \mathcal{E} : (\bar{s}_e, \bar{d}_e) = (g, h), P_e < c, t \leq L_e\}. \quad (19)$$

Display 20

$$e^\star = \min \mathcal{B}(g, h, t), \quad \mathcal{R}_{M_2}(g, h, t) = D_{e^\star}, \quad P_{e^\star} \leftarrow P_{e^\star} + 1. \quad (20)$$

Display 21

$$e^\star = \underset{e \in \mathcal{E}}{\text{lex arg min}}(B_e, e). \quad (21)$$

Display 22

$$\begin{aligned} L_{e^\star} &= \max\{t, B_{e^\star}\} + \gamma|G_{e^\star} - g| + \tau^L, \\ D_{e^\star} &= L_{e^\star} + \gamma|g - h| + \tau^U. \end{aligned} \quad (22)$$

Display 23

$$B_{e^\star} \leftarrow D_{e^\star}, \quad G_{e^\star} \leftarrow h, \quad (\bar{s}_{e^\star}, \bar{d}_{e^\star}) \leftarrow (g, h), \quad P_{e^\star} \leftarrow 1. \quad (23)$$

Display 24

$$\log Z \sim \mathcal{N}\left(-\frac{\sigma^2}{2}, \sigma^2\right), \quad \mathbb{E}[Z] = 1. \quad (24)$$

Display 25

$$\begin{aligned} L_{e^\star} &= \max\{t, B_{e^\star}\} + \gamma|G_{e^\star} - g|Z^{\text{rep}} + \tau^L Z^{\text{load}}, \\ D_{e^\star} &= L_{e^\star} + \gamma|g - h|Z^{\text{trav}} + \tau^U Z^{\text{unload}}. \end{aligned} \quad (25)$$

Display 26

$$Y_{km}(\xi) = C_{\max}(W^k; m, \xi) = \max_{j \in \mathcal{J}_n} t_{jkm}^C(\xi) - t_W. \quad (26)$$

Display 27

$$\mu_{qm}^{0.5} = \text{Med}_{\kappa \sim \nu_{q, \xi}} [Y_{km}(\xi)]. \quad (27)$$

Display 28

$$q^\star \in \arg \min_{q \in \mathcal{Q}} \max_{m \in \mathcal{M}_D} \mu_{qm}^{0.5}. \quad (28)$$

Display 29

$$\begin{aligned}
& \min_{y, \eta^Q} \eta^Q \\
& \text{s.t.} \quad \sum_{q \in \mathcal{Q}} y_q = 1, \\
& \quad \eta^Q \geq \sum_{q \in \mathcal{Q}} \mu_{qm}^{0.5} y_q, \quad m \in \mathcal{M}_D, \\
& \quad y_q \in \{0, 1\}, \quad q \in \mathcal{Q}, \\
& \quad \eta^Q \geq 0.
\end{aligned} \tag{29}$$

Display 30

$$\ell_{km} = C_{\max}(W^k; m), \quad k \in \mathcal{K}, \quad m \in \mathcal{M}_D. \tag{30}$$

Display 31

$$\min_{k \in \mathcal{K}} \max_{m \in \mathcal{M}_R} \ell_{km}. \tag{31}$$

Display 32

$$\begin{aligned}
& \min_{z, x, \eta^C} \eta^C \\
& \text{s.t.} \quad \sum_{k \in \mathcal{K}} z_k = 1, \\
& \quad x_o = \sum_{k \in \mathcal{K}} \delta_{ok} z_k, \quad o \in \mathcal{O}, \\
& \quad \sum_{o \in \mathcal{O}} x_o = n, \\
& \quad \eta^C \geq \sum_{k \in \mathcal{K}} \ell_{km} z_k, \quad m \in \mathcal{M}_R, \\
& \quad z_k \in \{0, 1\}, \quad k \in \mathcal{K}, \\
& \quad x_o \in \{0, 1\}, \quad o \in \mathcal{O}, \\
& \quad \eta^C \geq 0.
\end{aligned} \tag{32}$$

Unique inline expressions

1. t_W
2. $r_o \geq 0$
3. $t_W + r_o$
4. f^0
5. $f \in \mathcal{F} = \{1, \dots, F\}$
6. $o \in \mathcal{O}$
7. $a \in \mathcal{A} = \{1, \dots, A\}$
8. $e \in \mathcal{E} = \{1, \dots, E\}$
9. $k \in \mathcal{K} = \{1, \dots, K\}$
10. $q \in \mathcal{Q}$
11. $m \in \mathcal{M} = \{M_1, M_2, M_3\}$
12. F
13. A
14. E
15. c
16. n
17. K
18. $f^0 \in \mathcal{F}$
19. ι_o
20. o
21. $s_o, d_o \in \mathcal{F}$
22. $\gamma > 0$
23. $\tau^L, \tau^U > 0$
24. $\tau^P, \tau^D > 0$
25. $\alpha \in (0, 0.5)$
26. $\sigma > 0$
27. M_3
28. $1 \leq n \leq |\mathcal{O}|$
29. $k \in \mathcal{K}$
30. $W^k \subseteq \mathcal{O}$
31. δ_{ok}
32. W
33. $p_f(W)$
34. $2|W|$
35. f
36. $\mathbb{I}(\cdot)$
37. $\log$
38. C
39. $I \in [0, 1]$

40. $\bar{r}_W = |W|^{-1} \sum_{o \in W} r_o$
41. T
42. $T = 0$
43. I
44. $X \in \{C, I\}$
45. $Q_u(X)$
46. u
47. α
48. HH, HL, LH, LL
49. $\mathcal{Q} = \{q \in \{\text{HH, HL, LH, LL}\} : \mathcal{K}_q \neq \emptyset\}$
50. $\mathcal{Q} \neq \emptyset$
51. q
52. $\kappa \sim \nu_q$
53. W^κ
54. k
55. $j \in \mathcal{J}_n = \{1, \dots, n\}$
56. $\pi(W^k)$
57. m
58. ξ
59. H_a^{j-1}
60. G_a^{j-1}
61. a
62. $j-1$
63. $H_a^0 = t_W$
64. $G_a^0 = f^0$
65. o_j^k
66. g
67. $h \neq g$
68. t
69. $\mathbf{S}_m$
70. h
71. $\mathcal{R}_m(g, h, t)$
72. $\hat{t}$
73. $\mathbf{S}_m^+$
74. $a \neq a_j$
75. $H_a^j = H_a^{j-1}$
76. $G_a^j = G_a^{j-1}$
77. M_1
78. Ec
79. $v \in \mathcal{V}_1 = \{1, \dots, Ec\}$
80. B_v

81. G_v
82. $B_v = t_W$
83. $G_v = f^0$
84. $B_{v^*} \leftarrow \mathcal{R}_{M_1}(g, h, t)$
85. $G_{v^*} \leftarrow h$
86. M_2
87. e
88. B_e
89. G_e
90. $(\bar{s}_e, \bar{d}_e)$
91. L_e
92. D_e
93. P_e
94. $B_e = D_e = t_W$
95. $G_e = f^0$
96. $L_e = -\infty$
97. $P_e = 0$
98. $\mathcal{B}(g, h, t) \neq \emptyset$
99. $\mathcal{B}(g, h, t) = \emptyset$
100. $\mathcal{R}_{M_2}(g, h, t) = D_{e^*}$
101. $Z^{\text{rep}}, Z^{\text{load}}, Z^{\text{trav}}, Z^{\text{unload}}$
102. $\mathcal{R}_{M_3}(g, h, t) = D_{e^*}$
103. $\xi \in \Xi$
104. Ξ
105. κ
106. $t_{jkm}^C(\xi)$
107. y_q
108. $\{0, 1\}$
109. η^Q
110. $\mathbb{R}_+$
111. z_k
112. x_o
113. η^C
114. $m \in \mathcal{M}$
115. $\mathcal{M}_D = \{M_1, M_2\}$
116. $\mathcal{M}_R \subseteq \mathcal{M}_D$
117. $\mathcal{M}_R = \{m\}$
118. $\mathcal{M}_R = \mathcal{M}_D$
119. $\mathcal{M}_R$